# MEssentials: A Centralized Student Hub for UP Diliman's Department of Mechanical Engineering


**Presented by:**

*Krispy Little Potatoes*

Celestino, Lily Anne P.

Nario, Christian Kent N.

Sandig, Prince Sydric V.

**Commissioned by:**

Prof. Albert Jr V. Fibra

**Submitted to:**

Dr. Ligaya Leah L. Figueroa

# Customer Statement of Requirements

For our software engineering project in CS 191, under the guidance of Dr. Ligaya Leah L. Figueroa, we consulted Prof. Albert Jr V. Fibra, the Head of Student Affairs in UP Diliman's Department of Mechanical Engineering (DME). He introduced us to the Mechanical Engineering (ME) Representative and Mechanics Volcorps, a student-led initiative that works alongside the Engineering Student Council, Engineering Freshie Council, and the department to provide student services for their constituents. This includes, but is not limited to, the following initiatives:

- HonestME Store, an honesty store project where students can buy stationery, sanitary products, and snacks
- ME-kailangan, an item-borrowing initiative where students can temporarily borrow school supplies (calculators, ES1 kits), utilities (umbrellas, tools), and recreational items (board games)
- MEg-aral, an academic resource initiative focused on providing review materials for courses relevant to ME students
- ME Days, a recurring initiative focused on organizing recreational, academic, and career-related events for students

Aside from these projects, the Mechanics Volcorps also assist with student registration during the start of each semester.

Aside from information about student registration, which is disseminated through printed QR codes inside the IE-ME Building during the face-to-face registration period, most of the information about the Mechanics Volcorps' projects is announced on the ME Representatives and Mechanics Volcorps and ME Days Facebook pages, which can be accessed through the links below:

- https://www.facebook.com/mechenggreps
- https://www.facebook.com/UPMEDays

Said announcements mostly include introductions for new initiatives and updates on existing projects (ie: new reviewers released in MEg-aral, HonestME Store restocks, new borrowable equipment in MEg-kailangan, details about the next ME Day). Aside from announcements regarding initiatives, the page is also used for posting statements during holidays, updates on class suspensions and relevant current events, initiatives of other organizations in DME, and more. To further disseminate the information, links to these announcement posts are also sent in the official Messenger group chats for ME students, which, just like the Facebook page, should already be populated with chats on topics irrelevant to student services.

Aside from info dissemination, many of these initiatives are also reliant on external apps and websites. For example, the MEg-aral initiative uses Google Docs as both a navigation tool and platform for review materials, MEg-kailangan uses Google Forms to track item borrowing, and ME Days uses Google Sheets to track event attendance. Therefore, many posts on the Facebook page also end up

redirecting the students to yet another link, adding another layer of interaction for availing student services.

Another service that makes use of Google Forms is student registration, as it is used to both input their student information and add them to the registration queue. One can know when they're next in the queue by manually searching their student number on a spreadsheet in Google Sheets, another application, and checking whether it's their turn. However, this can be somewhat tedious, which is why volunteers from the Mechanics Volcorps help by calling whoever is next in the queue. The lack of an easy way to know one's queue status complicates the process by requiring manpower, which makes registration more inefficient than it should be.

It is apparent that there are many issues and inconveniences with the current way the Mechanics Volcorps handles their student services. First, information about their initiatives is confined to a Facebook page, which, by design, shows each post in chronological order, with the most recent at the top. This means important information about their initiatives, especially initiatives that have been released a long time ago, gets easily buried by other posts irrelevant to student services, making learning about it very inconvenient if you are a new ME student or you're not an avid follower of the page.

Another problem is the sheer amount of links and applications that are needed to redirect users to access the student services they need. After learning about the initiative from Facebook or Messenger, students then have to go

through an intermediate application like Google Docs, Forms, or Sheets. This is especially a problem for Google Forms, as they have to input their information, which is repetitive considering a lot of forms (ie: registration forms) ask for the same information (ie: name, student number) every time. This adds more clicks, apps, and browser tabs required just to interact with one service, making the process inconvenient. This is also a problem for the volunteers, because they have to use a lot of apps to make announcements and track metrics on their initiatives, making their digital workspace more bloated than it should be.

In worse cases, some of the current systems used to facilitate projects are inadequate in fulfilling their function. This is particularly true for the student registration process, as the current queue system is inconvenient for both the students and volunteers involved. In addition to this, the use of a publicly available spreadsheet to view one's queue status exposes information about other students, which shouldn't be necessary.

To solve these problems, our group proposes MEssentials, a centralized student hub that includes easy access to most of the Mechanics Volcorps and department initiatives, specifically the following:

- HonestME Store
- MEg-kailangan
- MEg-aral
- ME Days
- Student registration

This allows students to easily access all of the available initiatives of the volcorps in just one application. In particular, students should be able to do the following:

- Log in using their UP Mail
- Update their profile with relevant information (to be used for automatic form answering)
- View introductory information regarding all Mechanics Volcorps initiatives
- View HonestME Store stocks and get notifications of restocks
- View MEg-kailangan items available for borrowing, submit a borrowing request form, and get notified of their item return deadline
- View MEg-aral review materials and get notifications for new releases
- View information, get notifications, and register for upcoming ME Day events
- Register, view their queue number, and get notified when they’re next in the queue during student registration
- Provide feedback and suggestions for the application

Additionally, the volcorps volunteers should be able to easily make changes on the website, validate student form responses, and view administrator-exclusive metrics. In particular, they have the following additional privileges:

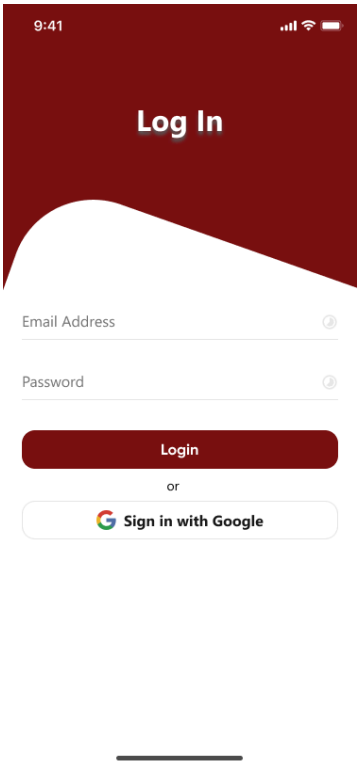
- Update student-viewable information for all Mechanics Volcorps initiatives
- Validate MEg-kailangan item borrows and returns
- Mark registered and walk-in students as attendants of ME Days events and monitor attendance statistics

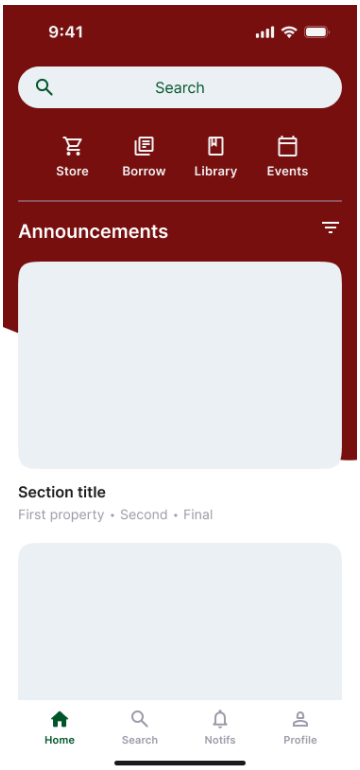
- View feedback and suggestions for the application

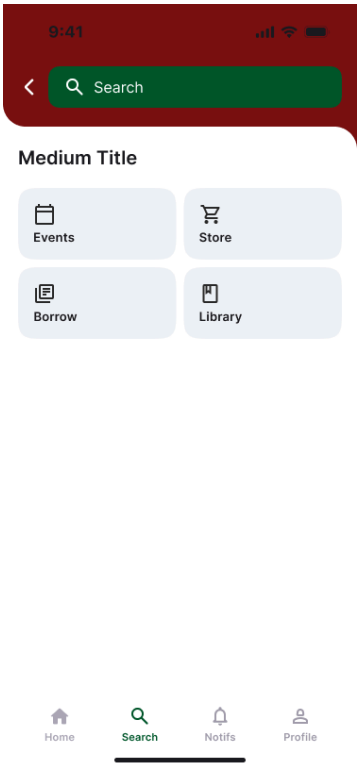
This project solves the previously mentioned problems of the current implementation of Mechanics Volcorps' initiatives: it collates all the information and updates on student services into one application, reduces the need for several external applications to facilitate said services, and fills in the lacking functionality of certain implemented systems.

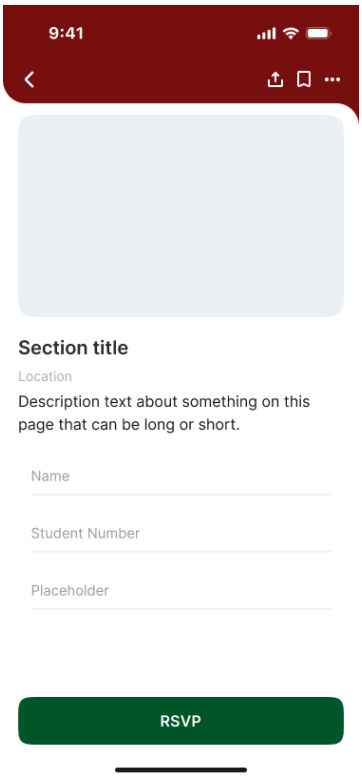
This project will be greatly beneficial to DME's constituents and the Mechanics Volcorps, as it will improve the functionality and convenience of the volcorps' student services, both for those who use them and those who provide them. In addition, this can also serve as a prototype for a general student service application, which may be used or serve as a reference for other departments and colleges in UP Diliman, and perhaps by other campuses and universities too.

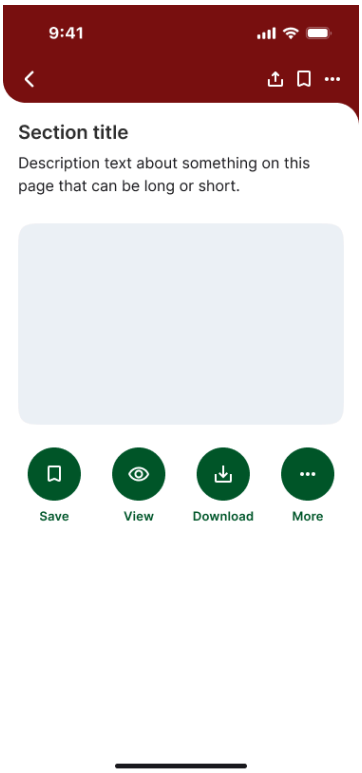
# Preliminary UI Design

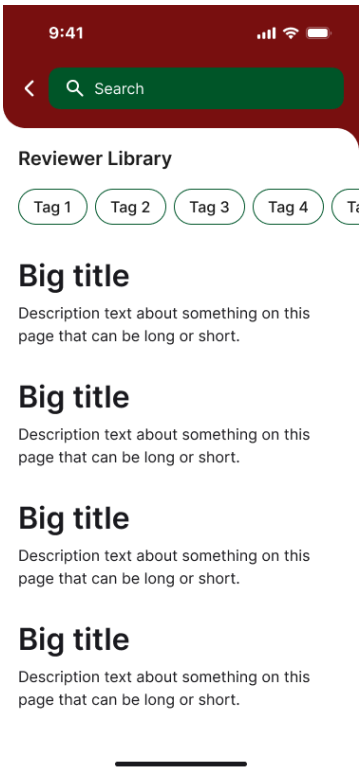


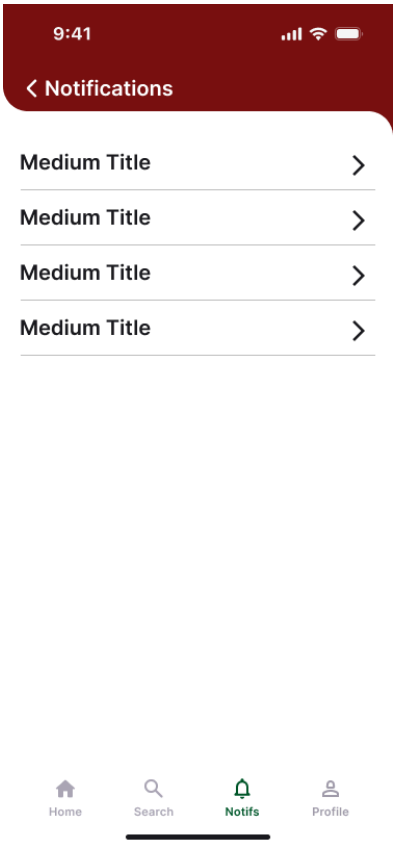



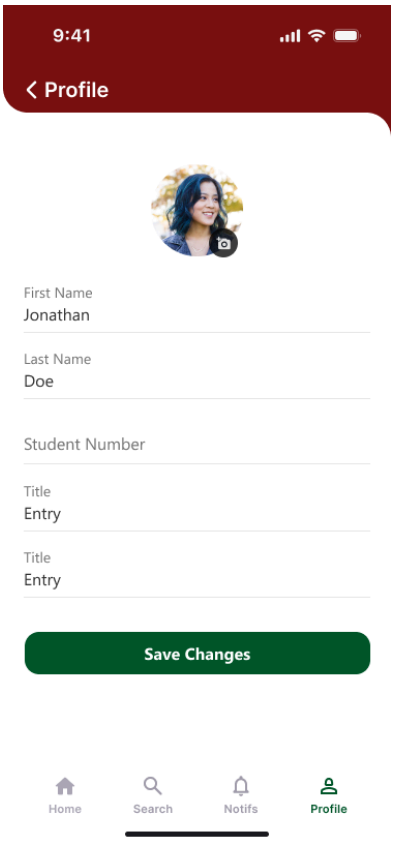



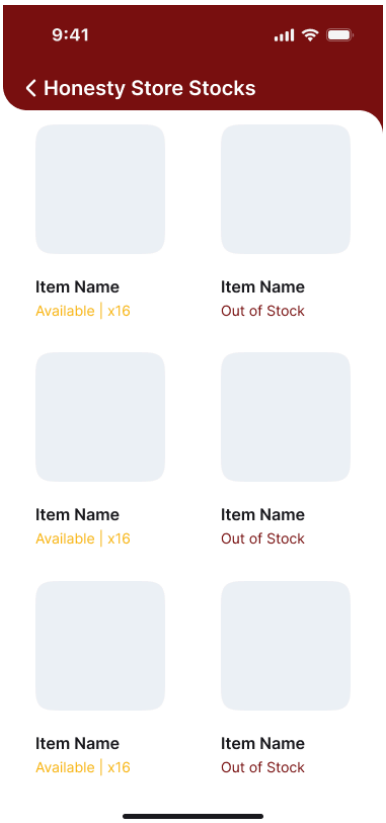



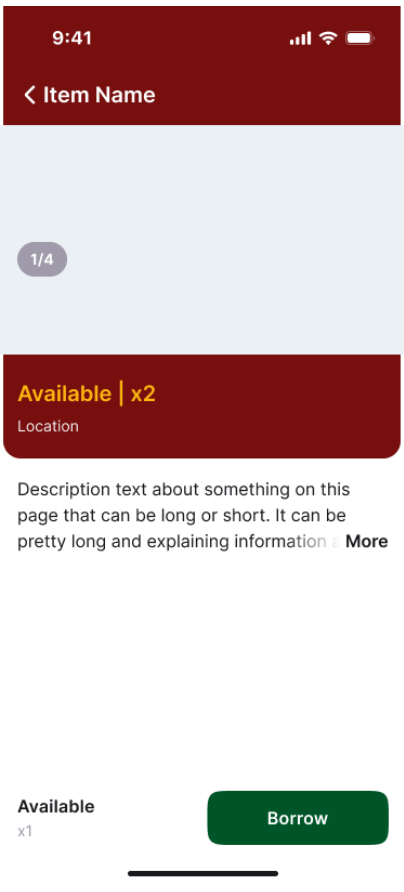



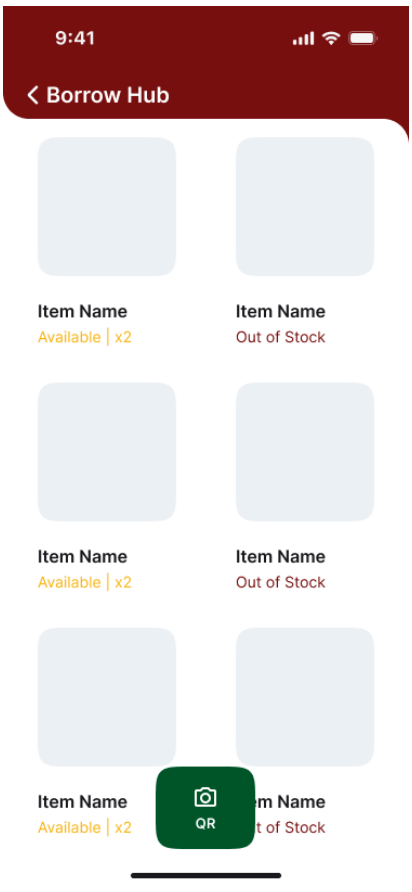



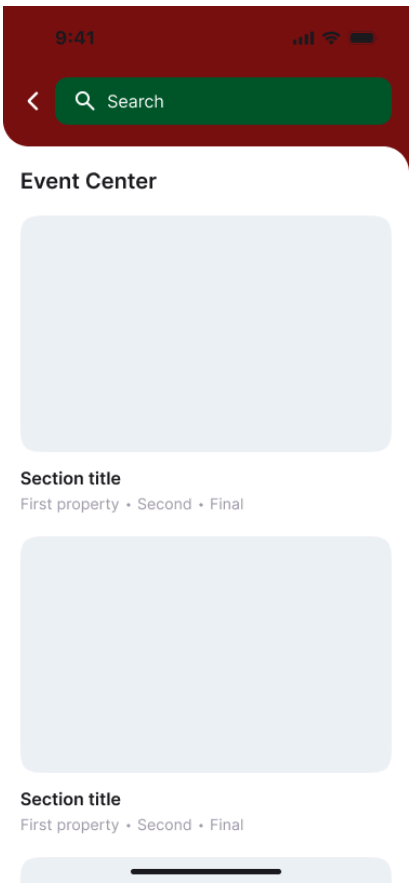

9:41
Notifications
Medium Title
Medium Title
Medium Title
Medium Title
Home
Search
Notifs
Profile


9:41
Profile
First Name
Jonathan
Last Name
Doe
Student Number
Title
Entry
Title
Entry
Save Changes
Home
Search
Notifs
Profile


9:41
Honesty Store Stocks
Item Name
Available | x16
Item Name
Out of Stock
Item Name
Available | x16
Item Name
Out of Stock
Item Name
Available | x16
Item Name
Out of Stock


9:41
Item Name
1/4
Available | x2
Location
Description text about something on this page that can be long or short. It can be pretty long and explaining information
More
Available
x1
Borrow


9:41
Borrow Hub
Item Name
Available | x2
Item Name
Out of Stock
Item Name
Available | x2
Item Name
Out of Stock
Item Name
Available | x2
QR
Out of Stock


9:41
Search
Event Center
Section title
First property • Second • Final
Section title
First property • Second • Final

**Navigational Paths**

Log In Screen ⇒ Home Page ⇒ {Feature of Choice} ⇒ Additional Information with Call to Action Buttons

# Plan of Work

## Estimated Cost and Resources

### Development Requirements

a. Human Resources:
   - 3 developers
   - In libraries around UP + remote work
   - Skills needed: app development and a little backend for the data

b. Environment:
   - Computers for development
   - Potentially 1 computer for a server to host the database for the app; if not, the cost of renting one

c. Reusable Software Resources
   - Database software

## FP-Based Estimation

Based on the metrics for WebApps given in the book, we will use function point-based estimation with the following changes:

1. Inputs are each input screen or form, each maintenance screen, each tab
2. Outputs are each static Web page, each dynamic Web page script, each report
3. Tables are each logical table in the database, each XML object
4. Interfaces are logical files
5. Queries are each externally published or use a message-oriented interface.

$$\mathbf{FP = count\ total \times [0.65 + 0.01 \times (F_i)]}$$

To get the count total, we filled out the table with the information:

1. External Inputs: Item QR code, Email, Password, Participation in Event, Queue Request
2. External Outputs: Reviewer Links, Honesty Store Inventory, Borrowing Inventory, Event Announcements, Honesty Store Receipt, Borrowing Return Date, Queue Position
3. External Inquiries: Item Inquiry, Reviewer Inquiry, Queue Position Inquiry, Event Inquiry, Borrow Date Update
4. Internal Logical Files: Reviewer Database, Honesty Store Database, Borrowing Database, Events Database, Queueing Management System

5. External Interface Files: Event Reminder, Borrowing Reminder, Queue Update

And that we want all of them to be simple.

| **Information Domain Value** | **Estimated Count** | **Weighting factor** | | | |
|---|---|---|---|---|---|
| | | **Simple** | **Average** | **Complex** | |
| External Inputs | 5 | 3 | 4 | 6 | 15 |
| External Outputs | 7 | 4 | 5 | 7 | 28 |
| External Inquiries | 5 | 3 | 4 | 6 | 15 |
| Internal Logical Files | 5 | 7 | 10 | 15 | 35 |
| External Interface Files | 3 | 5 | 7 | 10 | 15 |
| Count Total | | | | | 108 |

Getting value adjustment factors (0-5):

1. Does the system require reliable backup and recovery? **5**
2. Are specialized data communications required to transfer information to or from the application? **0**
3. Are there distributed processing functions? **0**
4. Is performance critical? **5**

5. Will the system run in an existing, heavily utilized operational environment? **4**
6. Does the system require online data entry? **5**
7. Does the online data entry require the input transaction to be built over multiple screens or operations? **3**
8. Are the ILFs updated online? **5**
9. Are the inputs, outputs, files, or inquiries complex? **0**
10. Is the internal processing complex? **0**
11. Is the code designed to be reusable? **3**
12. Are conversion and installation included in the design? **4**
13. Is the system designed for multiple installations in different organizations? **2**
14. Is the application designed to facilitate change and ease of use by the user? **5**

Therefore, **F_i = 41.**

From this, we can get that our **FP = 114**

## System Requirements

We aim to be able to run the software on any mobile device connected to the internet with the following minimum specifications:

- Storage: at least 1 GB of free space
- RAM: 1 GB and above
- OS: iOS 12.0 or later or Android 5.0 or later

## Schedules and Milestones

### Gantt Chart

| Task | Duration | Timeline | | | | |
|---|---|---|---|---|---|---|
| | | Week 1 | Week 2 | Week 3 | Week 4 | |
| Prototype App Development | 12 days | | | | | |
| Database Set-up | 5 days | | | | | |
| Testing (System) | 3 days | | | | | |
| Testing (User) | 3 days | | | | | |
| Polishing App | 7 days | | | | | |
| Migration | 4 days | | | | | |
| Testing (System) | 3 days | | | | | |
| Deployment | 1 day | | | | | |

## Project Metrics

Our team will be using the Defect Removal Efficiency (DRE) which will be used as an indicator of our quality control and assurance activities, making sure our program is free of as many errors as possible before delivery. The formula is defined as follows:

$$DRE = \frac{E}{E + D}$$

where:

- E is the number of errors found before delivery of the software to the end user.
- D is the number of defects found after delivery.

## Effort Breakdown

| Modules | Lily | Kent | Prince |
|---|---|---|---|
| Web App Modules | | | |
| Log In Screen | | | |
| Main Menu | | | |
| Header Interface | | | |
| Reviewer (student) | | | |
| Reviewer (volunteer) | | | |
| Honesty store (student) | | | |
| Honesty store (volunteer) | | | |
| Borrowing (student) | | | |
| Borrowing (volunteer) | | | |
| Registration Queue (student) | | | |
| Registration Queue (volunteer) | | | |

| | | | |
|---|---|---|---|
| Events (participant) | | | |
| Events (organizer) | | | |
| Database | | | |
| Setting up the software (accounts to be given to org) | | | |
| Migration to Database | | | |
| Reviewers | | | |
| Honestly Store Stocks | | | |
| Borrowing List + Users | | | |
| Queueing Management System | | | |
| Event Details | | | |
| UP Email Integration | | | |
| QR Scanner Integration | | | |
| Testing | | | |

# Project Risks

| Risks | Category | Probability | Impact | RMMM |
|---|---|---|---|---|
| High Traffic | TE/PS | 80% | 2 | Test the app for high traffic usage before launch. |
| QR Scanner Incompatibility | TE | 60% | 2 | Manual/Non-QR option |
| Low Student Usage | BU | 60% | 3 | Disregard |
| Event Overbooking | TE | 50% | 2 | Proper application of database transactions |
| Storage Exhaustion | PS/DE | 40% | 2 | Implement file size capping and/or cloud storage |

| | | | | |
|---|---|---|---|---|
| | | | | implementation |
| Delivery Delays | TE | 20% | 3 | Monitoring |
| Data Discrepancy | CU/BU | 20% | 3 | Monitoring |
| App Turnover | ST | 100% | 3 | Proper Documentation |

Impact values:

1—catastrophic

2—critical

3—marginal

4—negligible